NOT MEASUREMENT SENSITIVE

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MILITARY SPECIFICATION

PLANT, AGGREGATE, SECONDARY, WITH CONE CRUSHER, MOBILE, 75 TON PER HOUR CAPACITY

This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a complete mobile, electric powered, crushing and screening plant capable of producing not less than 75 tons of aggregate per hour suitable for use in cement or asphaltic concrete.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

QQ-S-781 - Strapping, Steel, and Seals.
SS-S-445 - Slag, Stone and Gravel, Crushed (for Bituminous Paving Mixtures).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A | FSC 3820 |

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TT-P-664 - Primer Coating, Synthetic, Rust-Inhibiting, Lacquer-Resisting.
TT-V-119 - Varnish, Spar, Phenolic-Resin.
UU-T-81 - Tags, Shipping and Stock.
MMM-A-260 - Adhesive, Water-Resistant, (for Sealing Waterproofed Paper).
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-1055 - Barrier, Material, Waterproof, Flexible.
PPP-P-40 - Packaging and Packing of Hand Tools; Tools and Tool Accessories for Power Driven, Metal and Woodworking Machinery.

MILITARY

MIL-C-104 - Crates, Wood: Lumber and Plywood Sheathed, Nailed and Bolted.
MIL-P-116 - Preservation, Method of.
MIL-B-121 - Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-V-173 - Varnish, Moisture-and-Fungus-Resistant (for Treatment of Communications, Electronic, and Associated Equipment.
MIL-C-3774 - Crates, Wood; Open 12,000- and 16,000-Pound Capacity.
MIL-V-13811 - Varnish, Waterproofing, Electrical Ignition.
MIL-E-16298 - Electric Machines Having Rotating Parts and Associated Repair Parts, Packaging of.
MIL-G-19826 - Generator Sets, Diesel Engine, Alternating Current, For Facilities Construction: 10 KW through 1000 KW (Not Stocked).
MIL-T-22085 - Tapes, Pressure-Sensitive, Adhesive, Preservation and Sealing.
MIL-G-28554 - Generator Sets, Mobile Electric Power and Supplemental Equipment: Packaging of.
MIL-C-52950 - Crates, Wood, Open and Covered.
MIL-P-53030 - Primer Coating, Epoxy, Water Reducible, Lead and Chromate Free.
MIL-B-81705 - Barrier Materials, Flexible, Electrostatic-Free, Heat Sealable.

STANDARDS

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-209 - Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.
MIL-STD-1186 - Cushioning, Anchoring, Bracing, Blocking, and Waterproofing, With Appropriate Test Methods.
MS51336 - Lunette and Coupler, Drawbar Ring.

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(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents. The following other Government document forms a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DEPARTMENT OF LABOR (DoL)
OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

DEPARTMENT OF TRANSPORTATION (DoT)

Federal Motor Carrier Safety Regulations

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

(Unless otherwise indicated, copies of other Government documents should be obtained from the contracting officer or as directed by the contracting officer.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DODISS shall be the issue of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM C 33 - Concrete Aggregates.
ASTM C 131 - Test for Resistance to Abrasion of Small Size Coarse Aggregate by Use of the Los Angeles Abrasion Machine.
ASTM C 170 - Test for Compressive Strength of Natural Building Stone.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA AB 1 - Molded Case Circuit Breakers and Molded Case Switches.
NEMA ICS 1 - General Standards for Industrial Controls and Systems.
NEMA ICS 2 - Industrial Control Devices, Controllers and Assemblies.
NEMA ICS 4 - Industrial Control Equipment and Systems, Terminal Blocks for.

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NEMA ICS 6 - Enclosures for Industrial Controls and Systems.
NEMA MG 1 - Motors and Generators.
NEMA PB 2 - Deadfront Distribution Switchboards.

(Application for copies should be addressed to the National Electrical Manufacturers Association, 2101 L Street, N.W., Washington, DC 20037.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 - National Electrical Code.

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269.)

NATIONAL LUBRICATING GREASE INSTITUTE (NLGI)

NLGI Lubricating Grease Guide.

(Application for copies should be addressed to the National Lubricating Grease Institute, 4635 Wyandotte Street, Kansas City, MO 64112.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE J534 - Lubrication Fittings.
SAE J551 - Performance Levels and Methods of Measurement of Electromagnetic Radiation From Vehicles and Devices (30 - 1000 MHz).

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

TIRE AND RIM ASSOCIATION, INC. (TRA)

TRA Yearbook.

(Application for copies should be addressed to the Tire and Rim Association, Inc., 3200 West Market Street, Akron, OH 44313.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, (except for related associated detail specifications, specifications sheets, or MS standards) the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Description. The plant, aggregate, secondary, with cone crusher, hereinafter called the aggregate plant, shall consist essentially of a cone crusher, vibrating screens, loading hopper, all electric power system, operating platform, and all accessories for a complete semitrailer mounted portable unit. When specified (see 6.2), the manufacturer shall provide a skid mounted diesel engine driven generator set (see 3.9.1).

3.2 Standard commercial product. The aggregate plant shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the aggregate plant being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.3 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.2.1 and 6.2).

3.4 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. Unless otherwise specified, none of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification.

3.5 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts.

3.6 Design and construction. The aggregate plant shall be so designed and constructed that normal adjustments, repair, and overhaul can be readily accomplished by means of general purpose tools with a minimum removal or disturbance of other elements of the aggregate plant. Ease of maintenance provisions shall insure operating clearances for facilitating maintenance and servicing. Where possible, intricate locking devices, controls, and threaded fastenings, that can be easily over-torqued by operators lacking feel through thick gloves or numbness, shall be avoided. Covers or plates that must be removed for component adjustment or for parts removal shall be equipped with substantial fastenings.

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3.6.1 Cone crusher. The cone crusher shall be of the vertical type with a mantel sized large enough to produce material requirements as stated in 3.10. The cone crusher shall be protected from damage due to overload, tramp iron, and other noncrushable material. The aggregate plant shall have a mechanical overload release and reset with no readjustment being required. The crushing members shall have replaceable wearing surfaces of alloy steel with not less than 11 percent manganese content. A hydraulic or mechanical means shall be provided to aid the adjustment for raising and lowering settings varying the crusher discharge opening. One extra set of crushing members, with resin, calking compounds, and other materials, if required, shall be furnished for the cone crusher. All moving parts of the cone crusher shall be lubricated by a force-feed system equipped with a pump, safety devices, oil temperature warning lights or alarm, low oil level and low oil pressure warning devices, instruments, flexible hoses, piping valves, and all necessary components for efficient operation of the cone crusher.

3.6.2 Vibrating screens. Vibrating screens shall be either the horizontal or inclined type, with not less than three decks designed to produce three products in addition to the oversize material which is routed to the cone crusher. Screen cloth shall be double crimp, standard or flattop oil-tempered steel with square openings, and adequate support shall be furnished to prevent sagging. Screens shall be readily replaceable. Screens shall be secured by an adjustable tensioning system. The vibrating mechanism shall be V-belt, electric-motor-driven, and shall consist of eccentric or unbalanced type drives for the horizontal type screen and one eccentric drive for the inclined type screen. A screen feed box or chute shall be provided to distribute the material on the screen. Master power panel for controlling and vibrating screens shall be operable from operator's platform. Two sets of screen cloths shall be furnished for the aggregate plant. Each set of screen cloths shall include the sizes shown in table I. The vibrating screen vibrations shall not be transmitted to the main structure. The receiving end of the vibrating screen shall be not greater than 14 feet in height. The receiving hopper shall be fabricated from rigidly reinforced steel plate.

3.6.3 Loading hopper. A loading hopper may be provided to control and direct flow of material from the vibrating screen to the cone crusher feed opening. The loading hopper shall be fabricated from reinforced steel plate.

3.6.4 Conveyors. Conveyors shall be used to transport stone and gravel within the aggregate plant. Conveyors shall be of the belted type in accordance with the manufacturer's standard practice to provide for smooth and efficient flow of material.

3.6.4.1 Belt-type conveyors. Belt-type conveyor shall be equipped as follows: Troughing type idlers or rollers shall be provided with spacing not greater than 4 feet between centers. Troughing idlers shall be designed to insure true running and self-centering of the conveyor belts. When required by manufacturer's design, special rolls designed to absorb shock shall be installed under the conveyor belt, below the receiving area. Return idlers shall be provided with spacing of not greater than 10 feet between centers.

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The width and running speed of all conveyor belts shall be such as to assure efficient movement of material, with a minimum of spillage and pileup at receiving points, when the aggregate plant is operating at full capacity. An accessible adjustment for maintaining proper belt tension shall be provided. Cleaning devices, capable of removing foreign material adhering to the pulley side of the belt, shall be provided. Conveyors which extend outside the basic dimensional limits of the unit shall be of the folding, retractable, or readily demountable type. Rubber coating on the carrying side of the belt shall be not less than 0.0625 inch thick and rubber coating on the bottom side shall be not less than 0.1875 inch thick. Each belt-type conveyor shall be driven from a separate electric or hydraulic motor. The capacity of the motors shall be sufficient to handle the loads encountered when the aggregate plant is operating at maximum capacity. Motors shall be positioned to transmit power directly or by V-belt to the head pulley via speed reducer. Inclined conveyors shall be designed to prevent rollback of the belt, in the event of power failure, if electric motors are used.

TABLE I. Vibrating screen cloth sizes.

Clear opening (minimum) (inches)	Wire diameter (inches)
2.50	0.50
2.00	0.4375
1.50	0.375
1.00	0.3125
0.75	0.25
0.50	0.192
0.375	0.162
0.25	0.135

3.6.5 Operator's platform. The operator's platform shall be centrally located to enable the operator to observe the aggregate plant's conveyor and crusher operation. Controls for aggregate plant operation shall be mounted on the platform for safe convenient operation of the unit by one operator. Access ladders, handrails, and guardrails shall be provided. Antiskid surfaces shall be provided for the deck, catwalks, and ladders. Platforms and catwalks shall be furnished to provide access to points requiring routine maintenance and adjustment. All guardrails around platform shall be demountable to reduce cube during transport.

3.6.6 Bearings. Antifriction or bronze bearings used on the aggregate plant, exposed to the elements, shall be provided with a dust-sealed device to prevent entry of abrasive materials.

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3.7 Semitrailer. The stress imposed on any structural component of the semitrailer shall be not greater than 60 percent of the yield point. The semitrailer for the aggregate plant shall be equipped with a dolly and shall be capable of being safely towed over the road at speeds of not less than 15 miles per hour (mph). The dolly drawbar shall be equipped with a lunette ring in accordance with MS51336. The angle of departure shall be not less than 25 degrees and projections below the main frame members shall be held to a minimum. The gross weight on any one axle shall be not greater than 20,000 pounds (lb); for tandem axle assembly, not greater than 40,000 lb; for triple axle assembly, not greater than 60,000 lb. Service brakes, lighting, reflectors, and associated equipment shall conform to DoT Motor Carrier Safety Regulations. The semitrailer shall be equipped with straight air service brakes controllable from the towing vehicle cab. All necessary air hose and standard air couplings for connecting to the brake system of a towing vehicle shall be provided. The service brake installation shall include a breakaway system capable of automatically applying the brakes in case the semitrailer is accidentally separated from the towing vehicle and shall be capable of maintaining the application for not less than 15 minutes. Parking brakes, or chock blocks securely chained to vehicle frame, shall be provided for each wheel. Chock block chains shall be of sufficient length to permit placing the blocks either ahead or behind the wheels. The blocks shall be equipped with snaphooks for storage. No less than six screw-jacks shall be provided on each aggregate plant. The jacks shall be capable of supporting each unit when operating fully equipped and fully loaded. Supports near the feed end of the aggregate plant and near the cone crusher shall be provided for blocking the aggregate plant during operation. The jacks shall retract sufficiently to preclude interference with the units normal travel capacities. Service brakes, lighting, reflectors, and associated equipment shall conform to DOT Motor Carrier Safety Regulations.

3.8 Tires and rims. The tire and rim ratings shall conform to the TRA Yearbook recommendations for the type and size of tires furnished. Tire and rim sizes shall be the same for all wheels. Tires shall be tube or tubeless type with highway tread. Tires shall be of rated capacity not less than equal to the load imposed on each tire, measured at each wheel at the ground. Tires shall be not less than 100 level quality. When tube type tires are furnished, inner tubes shall be heavy-duty type, and shall be of proper size for the tires furnished.

3.9 Electrical equipment. Electric motors shall conform to NEMA MG 1 polyphase induction, totally enclosed, fan-cooled type. The rated horsepower (hp) for each motor shall be equal to or greater than the hp requirements of the equipment driven under all operating conditions. The hp of the motors shall be such as to allow progressive starting of all electrically operated components. The electric motors shall contain provisions for operation on 230 Volt (V) or 460 V, 60 Hertz (Hz), 3 phase, alternating current from an outside source or when required (see 3.1), a skid mounted diesel engine driven generator set . A main distribution panel conforming to NEMA PB 2 shall provide circuit breakers for each electrical circuit, and a master circuit breaker that will disconnect the entire unit

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from the electric power supply. Circuit breakers shall conform to NEMA AB 1. A motor controller operated by momentary contact start and maintained contact stop pushbuttons shall be provided for control of each electric motor. Motor controllers shall conform to NEMA ICS 1, ICS 2, ICS 4, and ICS 6, type 3 enclosure. Motor control pushbuttons and an emergency panic pushbutton, which controls tripping of the master circuit breaker, shall be installed in an easily accessible location at the operator's platform and each control station for each electric motor used on the aggregate plant. Panic pushbutton enclosure shall be red. Enclosures for motors, motor controllers, distribution panel, pushbuttons, and other similar electrical apparatus shall be weatherproof and dustproof. Wiring of the aggregate plant shall be furnished, including interconnections between individual components and when connected to 60 Hz, 3 phase power source. A permanently legible wiring diagram shall be installed on or near the main distribution panel. The function of all circuit breakers and controls shall be permanently marked on their respective panels or enclosures. Conductors shall be color coded or suitably identified. Wiring shall be adequately protected from physical damage. The wiring design and practices shall meet the applicable requirements of NFPA 70, including grounding requirements.

3.9.1 Skid mounted diesel engine driven generator set. When required (see 3.1), the manufacturer shall provide a type II, style A, class 1 skid mounted diesel engine driven generator set (referred hereinafter as the generator set) conforming to MIL-G-19826. The generator set shall be capable of providing all power requirements at the voltages specified in 3.9 to operate all electrical motors and accessories of the aggregate plant under the loads specified in 3.10.

3.9.1.1 Electromagnetic interference suppression. When specified (see 6.2), the generator set shall meet the electromagnetic interference suppression limits of SAE J551.

3.10 Performance. The aggregate plant shall be capable of crushing any variety of rock having an average compressive strength of not less than 32,000 pounds per square inch when tested in accordance with ASTM C 170; shall have a percentage of wear not greater than 20 when tested in accordance with ASTM C 131, and a free moisture content of not greater than 6.50 percent. The aggregate plant shall be capable of crushing rock ranging in size from 6.0 inches to 3.0 inches cross section. The aggregate plant shall be capable of producing not less than a total of 100 tons per hour (TPH) from 2.50-inch minus, suitable for use in cement concrete, and not less than a total of 75 TPH from 0.75-inch minus, suitable for use in asphaltic concrete. The sizes and percentages of the crushed aggregates produced shall conform to the general requirements of ASTM C 33 for cement concrete, or SS-S-445 for asphaltic concrete.

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3.11 Safety. All rotating or reciprocating parts, and parts subjected to high operational temperature, that are of such nature or are so located as to be or become a hazard to operating or attending personnel, shall be adequately guarded or insulated to the extent necessary to eliminate the hazard. All platforms, steps, and handholds shall be provided for easy access to the operator's position and other work areas. The aggregate plant shall comply with OSHA regulations in effect at time of manufacture.

3.12 Identification plate. An identification plate will be furnished by the contracting officer for each aggregate plant. The contractor shall stamp all necessary data in the blank spaces of the plate provided for that purpose, and securely affix a plate to each aggregate plant in a conspicuous place with nonferrous screws, rivets, or bolts not less than 0.125-inch in diameter. The applicable nomenclature contained in the contract item description shall be placed in the top blank.

3.13 Servicing and adjusting. Prior to acceptance of the aggregate plant by the Government, the contractor shall service and adjust the aggregate plant for immediate operational use as required in the operator's manual. The servicing and adjusting shall include at least the following:

a. Inflation of all tires.
b. Adjustment of brakes.
c. Proper functioning of all lighting and electrical systems.
d. Wheel alinement.
e. Adjustment of engine to include tune-up.
f. Complete lubrication with grades of lubricants recommended for ambient temperature at the delivery point.
g. Cooling system filled to capacity with a clean solution of equal parts by volume of water and antifreeze (ethylene glycol).

3.14 Cleaning, treatment, and painting. Portions of the aggregate plant normally painted shall be cleaned, treated, and painted as specified herein. The color of the finish coat shall be as specified (see 6.2). Surfaces to be painted shall be cleaned and dried to ensure that they are free from contaminants such as oil, grease, welding slag and spatter, loose mill scale, water, dirt, corrosion product, or any other contaminating substances. As soon as practicable after cleaning, and before any corrosion product or other contamination can result, the surfaces shall be prepared or treated to insure the adhesion of the coating system. The painting shall consist of at least one coat of primer and one finish coat. The primer shall be applied to a clean, dry surface as soon as practicable after cleaning and treating. Painting shall be with manufacturer's current materials according to manufacturer's current processes and the total dry film thickness shall be not less than 2.5 mils over the entire surface. The paint shall be free from runs, sags, orange peel, or other defects.

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3.15 Instruction plates. The unit shall be equipped with instruction plates suitably located, describing any special or important procedures to be followed in operating and servicing the equipment. Plates shall be of a material which will last and remain legible for the life of the equipment. Plates shall be securely affixed to the equipment with nonferrous screws or bolts of not less than 0.125-inch diameter.

3.16 Toolbox(s). One or more toolbox(s) shall be provided. The toolbox(s) shall be large enough to store all tools required for field service or maintenance. The toolbox(s) shall be steel with a hinged lid and a fastener to keep the lid secure when vibrated. The toolbox(s) shall be mounted in a protected accessible location.

3.17 Stenciling. The gross weight of each demountable component of the unit shall be stenciled on each side of the component in such a manner as to be readily discernible to personnel handling the equipment. The total gross weight of the aggregate plant shall be stamped on the identification plate by the contractor.

3.18 Prime equipment accessories. Prime equipment shall be shipped with all required accessories and tools unless written deviation is received from the contracting officer.

3.19 Lifting and tiedown attachments. The unit shall be equipped with lifting and tiedown attachments. Lifting and tiedown attachments shall conform to type II or type III of MIL-STD-209. A nonferrous transportation plate shall be provided and mechanically attached to the unit. Transportation plates shall be inscribed with a diagram showing the lifting attachments and lifting slings, the capacity of each attachment, and the required length and size of each sling cable. A silhouette of the item furnished showing the center of gravity shall be provided on the transportation plate. Tiedown attachments may be identified by stenciling or other suitable marking. Tiedown marking shall show which attachments are intended for the tie down of the unit on the carrier when shipped. Lifting devices shall be designed to exclude the use of spreader bars.

3.20 Lubrication. Means for lubrication shall be in accordance with the manufacturer's standard practice. The lubricating points shall be easily visible and accessible. Hydraulic lubrication fittings shall be in accordance with SAE J534. Where use of high pressure lubricating equipment, 1,000 pound-force per square inch (psi) or higher, will damage grease seals or other parts, a suitable warning shall be affixed to the equipment in a conspicuous location.

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3.21 Fungus resistance. When specified (see 6.2), electrical components and circuit elements, including terminal and circuit connections, shall be coated with varnish conforming to MIL-V-173, except that:

a. Components and elements inherently inert to fungi or in hermetically sealed enclosures need not be coated.
b. Current-carrying contact surfaces, such as relay contact points, shall not be coated.

3.22 System of measurement. The dimensions used in this specification are not intended to preclude the use of the metric system of measurement in the fabrication and production of the material, individual parts, and the finished product, provided form, fit, and function requirements are satisfied.

3.23 Workmanship.

3.23.1 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the material. Corners shall be square and true. Flame-cutting, using tips suitable for the thickness of the metal, may be used instead of shearing and sawing. All bends shall be made by controlled means to insure uniformity of size and shape. Precautions shall be taken to avoid oveheating and heated metal shall be allowed to cool slowly. External surfaces shall be free from burrs, sharp edges and corners, except when sharp edges or corners are required or where they are not detrimental to safety.

3.23.2 Bolted connections. Boltholes shall be punched or drilled within manufacturer's design tolerances and shall have the burrs removed. Washers or lockwashers shall be provided, and all bolts, nuts, and screws shall be tight.

3.23.3 Riveted connections. Rivet holes shall be punched or drilled within manufacturer's design tolerances and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.23.4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

3.23.5 Castings. All castings shall be sound and free from patching, misplaced coring, warping, or any other defect which reduces the castings ability to perform its intended function.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practic to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. The first article inspection shall be performed on one aggregate plant when a first article is required (see 3.3 and 6.2). This inspection shall include the examination of 4.3 and the tests of 4.4. The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.3, the tests of 4.4, and the packaging inspection of 4.5.

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4.3 Examination. Each aggregate plant shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.4 Tests. The component units shall be arranged in the order as shown in the approved arrangements plan of the aggregate plant and the piping and accessories shall be connected to the component units for compatibility and subjected to test in accordance with 4.4.1 through 4.4.4. Nonconformance to the requirements as specified in section 3 shall constitute failure and cause for rejection.

4.4.1 Test conditions. Prior to being tested, the aggregate plant shall be lubricated. Oils shall be those designated for use in the ambient temperature at which the tests are being performed. During all aggregate plant tests, the motors shall be run at normal operating speeds. When the generator set is required (see 3.1), the aggregate plant shall be disconnected from the normal power supply and connected to the generator set. The generator set shall supply all the necessary electrical power to operate all components and accessories of the aggregate plant at normal operating speeds. As services and adjustments are performed, the aggregate plant shall be checked for ease of maintenance.

4.4.2 Operational test. The aggregate plant shall be completely assembled, adjusted, lubricated, or otherwise serviced for operation. The motors shall be started and subjected to a warmup period as recommended by the manufacturer. The aggregate plant shall be given a run-in test and all controls operated a sufficient number of times to ascertain that the components and mechanisms actuated by the controls operate promptly, fully, and without restriction of malfunction. When the generator set is required (see 3.1), the aggregate plant shall be disconnected from the normal power supply and connected to the generator set. The operational test shall then be repeated, using the generator set as the sole power supply for the aggregate plant. Each wheeled unit shall be towed a distance of not less than 10 miles at an average speed of not less than 15 mph over unpaved road. Failure to pass any phase of this test shall be cause for rejection.

4.4.3 Electromagnetic interference suppression. When specified (see 6.2), the generator set shall be tested in accordance with SAE J551 to meet the requirements of 3.9.1.1.

4.4.4 Lifting and tying down attachment tests. When lifting and tying down attachments are required, they shall be tested to conform to 3.19.

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4.5 Packaging inspection. The preservation, packing, and marking of the item shall be inspected to verify conformance to the requirements of section 5.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or commercial as specified (see 6.2).

5.1.1 Level A. The generator set, when provided (see 3.1), shall be preserved as specified in MIL-G-28554. The aggregate plant shall be preserved as specified in the following paragraphs.

5.1.1.1 Methods of preservation. Cleaning processes, drying procedures, preservatives, and methods of preservation specified in the following paragraphs are listed in MIL-P-116 and shall conform to the requirements of MIL-P-116 and any applicable specifications.

5.1.1.2 Disassembly. Disassembly shall be the minimum necessary to protect parts subject to damage or loss, and to accomplish reduction in cube. Removed bolts, nuts, pins, screws and washers shall be reinstalled in mating parts and secured to prevent their loss.

5.1.1.3 Matchmarking. Parts removed and mating parts on the equipment and attachments shall be matchmarked to facilitate reassembly. Parts and accessories removed, and mating parts on the equipment, shall be identified with weatherproof tags, conforming to UU-T-81, attached to mating parts and locations. Markings shall be applied to the tags with a waterproof material.

5.1.1.4 Cleaning and drying. Prior to the application of preservative compounds or paint, surfaces shall be cleaned by process C-1 and dried by any applicable procedure of MIL-P-116.

5.1.1.5 Unprotected surfaces. Unprotected exterior metal surfaces requiring the application of a contact preservative in accordance with MIL-P-116 and not specifically provided for herein shall be preserved as follows:

5.1.1.5.1 Unfinished (not machined) surfaces. Unfinished exterior metal surfaces shall be coated with type P-1 preservative.

5.1.1.5.2 Machined surfaces. Exposed machined surfaces shall be coated with type P-6 or P-11 preservative and wrapped or covered, as applicable, with barrier material conforming to MIL-B-121, type I, grade A, class 2. The material shall be secured in place with waterproof tape.

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5.1.1.6 Drive belts and pulleys. Drive belts shall be removed or released from tension. Removed belts shall be preserved method IC-1 or IC-3. Unpainted surfaces shall be coated with primer conforming to TT-P-664 or MIL-P-53030. A weatherproof tag, conforming to UU-T-81, shall be attached in a conspicuous location indicating: "Belts have been (removed or released from tension). (Install or tension) prior to operation." Markings shall be applied to the tags with a waterproof material.

5.1.1.7 Hydraulic systems. The hydraulic fluid supply tanks shall be filled to the operating level with hydraulic fluid required for operation. The pistons shall be retracted as far as practicable into the cylinders and secured. When the pistons cannot be fully retracted, the exposed portions of the piston rods (ramshafts) shall be coated with type P-6 or P-11 preservative and the coated surfaces wrapped or covered with barrier material conforming to MIL-B-121, type I, grade A, class 2, extending the wraps approximately 2 inches onto the ram cylinders. The wraps shall be secured in place with waterproof tape. When the pistons can be fully retracted, any remaining uncoated surfaces of the piston rods shall be coated with type P-1 preservative, with no wrapping required. The hydraulic control valves shall be secured in the neutral position and preserved as specified herein for piston rods. Hoses shall not be disconnected. A weatherproof tag, conforming to UU-T-81, shall be attached to the control lever indicating: "The hydraulic supply tank is filled to the operating level with fluid required for operation. Do not drain." Markings shall be applied to the tag with a waterproof material.

5.1.1.8 Electrical motors. Preserve level A by the alternate method of MIL-E-16298.

5.1.1.9 Electrical modules. Circuitry modules, printed circuit boards, and panel boards that are not hermetically sealed should be removed, if possible, and packaged method II. If the complexity of the circuitry will not allow removal, seal area with a waterproof barrier conforming to MIL-B-81705 or tape conforming to MIL-T-22085, type II or PPP-T-60, type IV.

5.1.1.10 Conveyor belts. Metal facings shall be coated with varnish conforming to TT-V-119 or MIL-V-13811. Each belt shall be completely rolled with a tube or piece of round wood placed in each end fold of the belt to prevent sharp bends. Prior to rolling, belt surfaces shall be dusted with powdered talc to prevent surfaces from adhering together. When required, to hold belt in compact roll, strapping conforming to QQ-S-781, class 1, type I, finish B, shall be placed around the roll. Suitable protective cushioning shall be placed under the strapping, and the strapping not tensioned so tight as to damage the belting. Each rolled belt shall be wrapped with waterproofed barrier conforming to PPP-B-1055, class E-1 or E-2, with all laps, seams, and folds sealed with adhesive conforming to MMM-A-260.

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5.1.1.11 Lubrication. Lubricate all grease fittings with type P-11 preservative to include universal joints, chassis suspension and steering linkage, fifth wheels, propeller shaft, and parking brake cable. Wheel bearings requiring a NLGI No. 2 grade grease shall be lubricated with type P-11 preservative. Lubricate control mechanisms, linkage, hinges, fasteners, and all pivot points with P-10.

5.1.1.12 Brakes. Clean all surfaces as required to remove contaminants. Coat unpainted exterior surfaces of rods, levers, cables, clevises, linkage, and associated parts with type P-1 preservative. Remove brake drum covers and spray enclosed metallic components with primer TT-P-664 or MIL-P-53030. Spray coat brake drum friction surfaces with a very thin coat of primer. Coat friction surfaces of cams, adjusting screws, and anchor pins with type P-6 or P-7 preservative. Back off all brake shoe adjusters for maximum clearance between brake drum and brake shoes. Do not apply coating of any type to contact surfaces of electric brakes. Adjust brake shoes to provide maximum clearance to drum surfaces and lined surfaces. In a conspicuous location in the operator's compartment, attach a weatherproof warning tag, conforming to UU-T-81, marked, "Brakes preserved. Do not apply brakes when vehicle is being moved. Use tow bar or similar arrangement for moving vehicle. Adjust brakes before placing equipment into operation." Markings shall be applied with waterproof material.

5.1.1.13 Rubber tires. Tires mounted on the wheels (road tires) of the vehicle shall be inflated to 10 pounds above pressure value recommended for maximum load. Tires mounted on rims (spare tires) shall be inflated to two-thirds of the specified operating pressure.

5.1.1.14 Maintenance tools. Maintenance tools shall be preserved in accordance with level A preservation and packaging requirements of PPP-P-40.

5.1.1.15 Technical publications. Technical publications for each piece of equipment shall be preserved method IC-1 or IC-3.

5.1.1.16 Consolidation. Tool boxes shall be utilized for the consolidation of loose components, tools, and publications. Tool boxes shall be secured to prevent pilferage. The remainder of the components that will not fit in the tool boxes shall be consolidated, along with disassembled components, in boxes conforming to PPP-B-601, overseas type or PPP-B-621, class 2. The contents shall be cushioned, blocked, and braced to prevent movement in accordance with MIL-STD-1186. Boxes shall be secured to the equipment with appropriate strapping. Arrangement and location on the equipment shall be such so as not to increase cubage or interfere with lifting or mobility of the equipment.

5.1.2 Commercial. The aggregate plant and the diesel engine driven generator set, when provided (see 3.1), shall be preserved in accordance with the contractor's standard commercial practice in a manner to prevent deterioration and damage.

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5.2 Packing. Packing shall be level A or commercial as specified (see 6.2).

5.2.1 Level A. The generator, when provided (see 3.1), shall be packed as specified in MIL-G-28554. Packing for the aggregate plant shall be as specified in the following paragraphs.

5.2.1.1 Basic plant (except conveyors). The basic aggregate plant shall be prepared for mobile shipment; openings shall be sealed to prevent the entry of free falling water. Large disassembled components that cannot be consolidated, packaged, or secured to the basic aggregate plant shall be packed in crates conforming to MIL-C-52950 or MIL-C-3774. The contents shall be anchored, blocked, and braced in accordance with the appendix to the crate specification.

5.2.1.2 Portable conveyors. Portable conveyors shall be shipped knocked down to the most practicable extent. The structural components shall be nested, arranged, and secured with bolts, or steel strapping, in combination with suitable wood blocking or battens, as required to form nonshifting bundles. The strapping shall be 0.035 inch by 1.25 inch strapping conforming to QQ-S-781, class 1, type I, finish B. The strapping shall be stapled to any wood blocking or battens provided. Where it is not practicable to disassemble the equipment to the extent required for suitable bundling, the sections shall be secured to skid bases conforming to the skid-type base of MIL-C-104, style A crates, and shipped as individual assemblies. The electric motors, conveyor belts, bolts, nuts, washers, and any other small components requiring the protection of a container for safe convenient handling shall be packed in containers as specified for disassembled components. Where skid bases are used, consideration shall be given to consolidating and securing any bundled or boxed components on the skid bases.

5.2.2 Commercial. The aggregate plant and the generator set, when provided (see 3.1), shall be prepared for shipment in a manner which will insure arrival at destination in a satisfactory condition. Preparation for delivery shall comply with applicable carrier rules and regulations.

5.3 Marking. Marking shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The aggregate plant covered by this specification is intended for use in crushing and screening aggregate to a uniform size.

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6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. When a type II, style A, class 1 skid mounted diesel engine driven generator set conforming to MIL-G-19826 is required (see 3.1 and 3.9.1).
d. When first article sample and inspection is required (see 3.3 and 4.2.1).
e. When electromagnetic interference suppression is required for the generator set (see 3.9.1.1).
f. Color of finish coat (see 3.14).
g. When treatment for fungus resistance is required (see 3.21).
h. When electromagnetic interference suppression testing is required for the generator set (see 4.4.3).
i. Level of preservation and level of packing (see 5.1 and 5.2).

6.3 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL) and invokes the provisions of DOD Federal Acquisition Regulations (FAR) Supplement 27.410-6, the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL (DD Form 1423) incorporated into the contract. When the provisions of DOD FAR 27.410-6 are not invoked, the data should be delivered in accordance with the contract requirements.

6.4 First article. When a first article inspection is required, the item should be a first production item, or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one complete aggregate plant unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first article. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.5 Subject term (key word) listing.

Aggregate plant
Cone crusher
Mobile
Secondary

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6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Navy - YD

Preparing activity:
Navy - YD

Project 3820-N028

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-82015C(YD)	PLANT, AGGREGATE, SECONDARY, WITH CONE CRUSHER, MOBILE, 75 TON PER HOUR CAPACITY

3a. NAME OF SUBMITTING ORGANIZATION

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

b. ADDRESS *(Street, City, State, ZIP Code)*

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional

b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional

c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional

8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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